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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

3086656949

CHEMISTRY **5070/21**

Paper 2 Theory **October/November 2011**

1 hour 30 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Section A
Answer **all** questions.
Write your answers in the spaces provided in the Question Paper.

Section B
Answer any **three** questions.
Write your answers in the spaces provided in the Question Paper.

A copy of the Periodic Table is printed on page 20.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
B6	
B7	
B8	
B9	
Total	

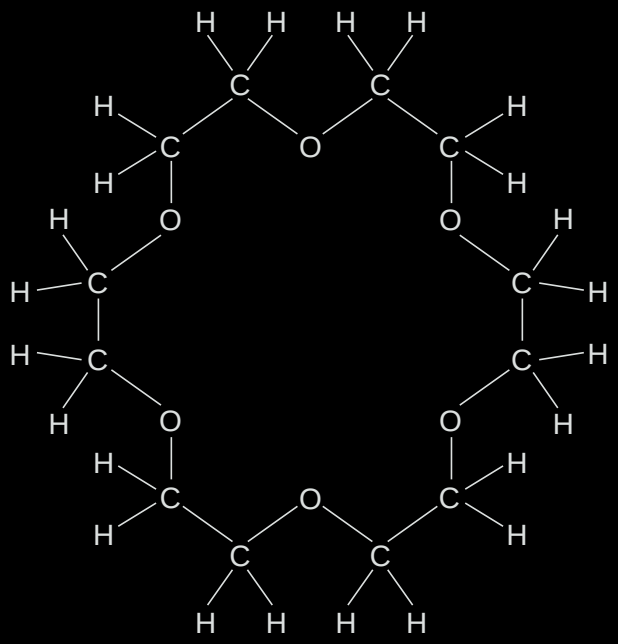
This document consists of **17** printed pages and **3** blank pages.

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Write the empirical formula for this crown ether.

..... [1]

(b) When sodium reacts with crown ethers it forms Na^+ and Na^- ions.
Draw the structure of an Na^- ion.
Show all the electrons.

[1]

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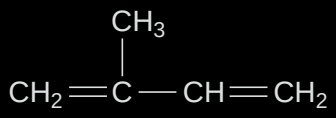
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isoprene

(i) Isoprene is an unsaturated compound.

What do you understand by the term *unsaturated*?

..... [1]

(ii) What would you observe when excess isoprene is added to aqueous bromine?

..... [1]

(d) In many plants, the alkene ethene promotes the ripening of fruits.

(i) Write the general formula for an alkene.

[1]

(ii) Draw the structure of an alkene containing four carbon atoms.
Show all atoms and bonds.

[1]

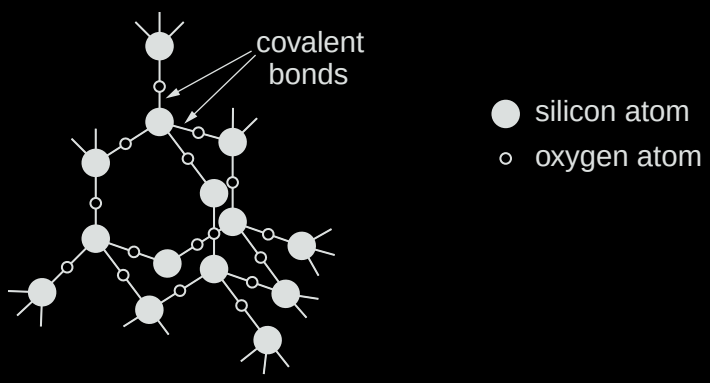
(iii) Describe how ethanol can be formed from ethene, stating the necessary reaction conditions.

.....

..... [2]

[Total: 14]

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Suggest why silicon dioxide

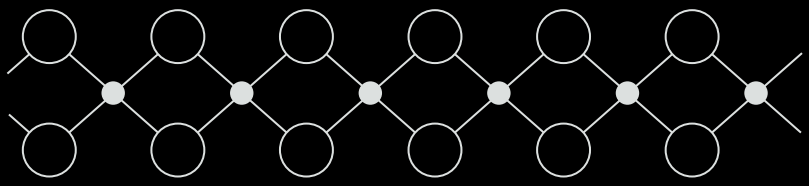
does not conduct electricity, ...

..... [1]

is hard. .....

..... [1]

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Deduce the empirical formula for palladium chloride.

..... [1]

(d) Sodium chloride has an ionic structure.
Explain why sodium chloride conducts electricity when molten but does not conduct electricity when in the solid state.

.....

.....

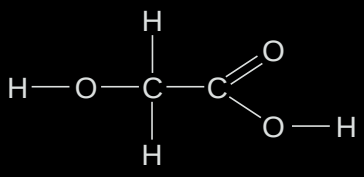
..... [2]

[Total: 9]

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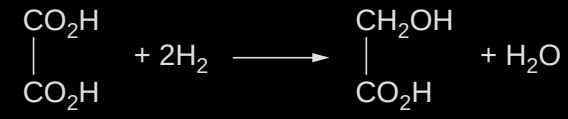
(a) Name the two functional groups present in glycollic acid.

..... and ..... [1]

(b) Glycollic acid undergoes similar reactions to ethanoic acid. Complete the equation for the reaction of glycollic acid with sodium carbonate.

$.....HOCH_2CO_2H + Na_2CO_3 \rightarrow$ ..... + ... + ... [2]

(c) Glycollic acid can be prepared from oxalic acid.



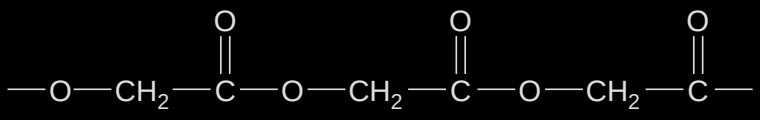
oxalic acid glycollic acid

How does this equation shows that oxalic acid has been reduced?

.....

..... [1]

(d) Glycollic acid polymerises to form poly(glycollic acid). The diagram shows a section of this polymer.



(i) Is poly(glycollic acid) an addition polymer or a condensation polymer? Give a reason for your answer.

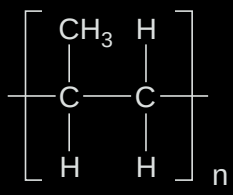
.....

..... [1]

(ii) Name another polymer with the same linkage as poly(glycollic acid).

..... [1]

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Draw the structure of the monomer used to make poly(propene).

[1]

[Total: 10]

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DATA SHEET
The Periodic Table of the Elements

Group																	
I	II										III	IV	V	VI	VII	0
							1 H Hydrogen 1									4 He Helium 2
7 Li Lithium	9 Be Beryllium 4										11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium	24 Mg Magnesium 12										27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K otassium	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	64 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
85 Rb Rubidium	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium	137 Ba Barium 56	139 La Lanthanum 57 *	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	209 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium	226 Ra Radium 88	227 Ac Actinium 89 †															

140 Ce Cerium 58	141 Pr Praseodymium 59	144 Nd Neodymium 60	147 Pm Promethium 61	150 Sm Samarium 62	152 Eu Europium 63	157 Gd Gadolinium 64	159 Tb Terbium 65	162 Dy Dysprosium 66	165 Ho Holmium 67	167 Er Erbium 68	169 Tm Thulium 69	173 Yb Ytterbium 70	175 Lu Lutetium 71
232 Th Thorium 90	231 Pa Protactinium 91	238 U Uranium 92	237 Np Neptunium 93	244 Pu Plutonium 94	243 Am Americium 95	247 Cm Curium 96	247 Bk Berkelium 97	251 Cf Californium 98	252 Es Einsteinium 99	257 Fm Fermium 100	258 Md Mendelevium 101	259 No Nobelium 102	260 Lr Lawrencium 103

8–71 Lanthanoid series
90–103 Actinoid series

y

a
X
b

a = relative atomic mass
X = atomic symbol
b = atomic (proton) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

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